MAT1575, Classwork8, Fall2026

3.2 Trigonometric Integrals (p.241-249)

1. *Case*1: Integrating Products and Powers of $\sin x$ and $\cos x$: $\int \sin^j(x)\cos(x)\,dx$ or $\int \cos^j(x)\sin(x)\,dx$

Evaluate $\int \cos^3(x)\sin(x)\,dx$ u-sub

$u=\cos(x)$, $du=-\sin(x)dx$, $-du=\sin(x)dx$

$= \int u^3(-du) = -\int u^3\,du = -\frac{u^4}{4}+C = -\frac{\cos^4(x)}{4}+C$

2. *Case*2: Integrating Products and Powers of $\sin x$ and $\cos x$: $\int \cos^j(x)\sin^k(x)\,dx$ **where k is odd.** u-sub

Evaluate $\int \cos^2(x)\sin^3(x)\,dx = \int \cos^2(x)\sin^2(x)\sin(x)\,dx = \int \cos^2(x)(1-\cos^2(x))\sin(x)\,dx$

$u=\cos(x)$, $du=-\sin(x)dx$, $-du=\sin(x)dx$

$= \int u^2(1-u^2)(-du) = -\int u^2(1-u^2)\,du$

$= -\int u^2-u^4\,du = -\left[\frac{u^3}{3}-\frac{u^5}{5}\right]+C = -\frac{\cos^3(x)}{3}+\frac{\cos^5(x)}{5}+C$

$\left(=\int u^4-u^2\,du\right)$

3. *Case*3: Integrating an Even Powers of $\sin x$ or $\cos x$: $\int \sin^k x\,dx$ or $\int \cos^k x\,dx$ **where k is even.** $\sin^2(x)=\frac{1-\cos(2x)}{2}$

Evaluate $\int \sin^2(x)\,dx = \int \frac{1-\cos(2x)}{2}\,dx = \int \frac{1}{2}-\frac{\cos(2x)}{2}\,dx$

$= \frac{x}{2}-\left(\frac{\sin(2x)}{4}\right)+C$

4. *Problem-Solving Strategy*: **Integrating Products and Powers of sin *x* and cos *x***

To integrate $\int \cos^j(x)\sin^k(x)\,dx$, we have 3 cases:

(a) k is odd: rewrite $\sin^k(x)=\sin^{k-1}(x)\cdot\sin(x)$ and use $\sin^2(x)=1-\cos^2(x)$ and $u=\cos(x)$

(b) j is odd: rewrite $\cos^j(x)=\cos^{j-1}(x)\cdot\cos(x)$ and use $\cos^2(x)=1-\sin^2(x)$ and $u=\sin(x)$

(c) both k and j are even: use $\sin^2(x)=\frac{1-\cos(2x)}{2}$ and $\cos^2(x)=\frac{1+\cos(2x)}{2}$, then go back to (a) or (b)

5. *Case4*: Integrating Products of Sines and Cosines of Different Angles $\sin(ax)$, $\sin(bx)$, $\cos(ax)$, $\cos(bx)$.

Evaluate $\int \sin(5x)\cos(3x)\,dx$. *Hint*: use sum of angle formula $\sin(ax)\cos(bx)=\frac{1}{2}\left(\sin((a-b)x)+\sin((a+b)x)\right)$

$= \int \frac{\sin(8x)+\sin(2x)}{2}\,dx = \int \frac{\sin(8x)}{2}+\frac{\sin(2x)}{2}\,dx$

$= -\frac{\cos(8x)}{8\cdot 2}-\frac{\cos(2x)}{2\cdot 2}+C$

The other two: $\sin(ax)\sin(bx)=\frac{1}{2}\left(\cos(a-b)x-\cos(a+b)x\right)$, and $\cos(ax)\cos(bx)=\frac{1}{2}\left(\cos(a-b)x+\cos(a+b)x\right)$

6. Evaluate $\int \cos^8(x)\sin^5(x)\,dx$ u-sub.

$= \int \cos^8(x)\sin^2(x)\sin^2(x)\sin(x)\,dx$

$u=\cos(x)$, $du=-\sin(x)dx$

$= \int \cos^8(x)\cdot(1-\cos^2(x))(1-\cos^2(x))\sin(x)\,dx$

$= \int u^8(1-u^2)(1-u^2)(-du)$

$= \int -u^8(1-2u^2+u^4)\,du = \int -u^8+2u^{10}-u^{12}\,dx$

$= -\frac{u^9}{9}+\frac{2u^{11}}{11}-\frac{u^{13}}{13}+C$

$= -\frac{\cos^9(x)}{9}+\frac{2\cos^{11}(x)}{11}-\frac{\cos^{13}(x)}{13}+C$

7. *Review*: The known integrals involving $\tan(x)$ and $\sec(x)$.

$\int \sec^2(x)\,dx =$	$\int \sec(x)\tan(x)\,dx =$	$\int \tan(x)\,dx =$	$\int \sec(x)\,dx =$

8. Evaluate $\int \sec^5(x)\tan(x)\,dx$ by using 7. directly.

9. *Case*1: Integrating $\int \tan^k(x)\sec^j(x)\,dx$ **when *j* is even** using $\sec^2(x) =$ and let $u =$

Evaluate $\int \tan^6(x)\sec^4(x)\,dx$

10. *Case*2: Integrating $\int \tan^k(x)\sec^j(x)\,dx$ **when *k* is odd** using $\tan^2(x) =$ and let $u =$

Evaluate $\int \tan^5(x)\sec^3(x)\,dx$

11. *Case*3: Integrating $\int \tan^k(x)$ when ***k* is odd**, **and *k* ≥ 3** using $\tan^2(x) =$ and let

Evaluate $\int \tan^3(x)\,dx$

12. *Case*4: Integrating $\int \sec^j(x)\,dx$ **when *j* is odd and *j* ≥ 3** using integration by parts and $\tan^2(x) = \sec^2(x) - 1$

Evaluate $\int \sec^3(x)\,dx$

13. *Problem-Solving Strategy*: **Integrating Products and Powers of tan *x* and sec *x***

To integrate $\int \sec^j(x)\tan^k(x)\,dx$, we have 4 cases:

(a) j is positive even:

(b) k is odd and $j \geq 1$:

(c) $k \geq 3$ is odd and $j = 0$:

(d) k is even and j is odd:

14. Reduction Formulas:

$$\int \sec^n(x)\,dx =$$

$$\int \tan^n(x)\,dx =$$

(which is the same as 13 (c))